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WaCPro: An Open-Source Application for Waveform and Crossbar Programming in Nanotechnology Research

IOANNIS K. CHATZIPASCHALIS [1,2] (Student Member, IEEE), PANTELIS FRAIDAKIS [1], GEORGIOS K. KLEITSIOTIS [1] (Graduate Student Member, IEEE), IOANNIS TOMPRIS [1] (Student Member, IEEE), ATHANASIOS PASSIAS [1] (Graduate Student Member, IEEE), EMMANOUIL STAVROULAKIS [1] (Member, IEEE), EVANGELOS TSIPAS [1] (Graduate Student Member, IEEE), THEODOROS PANAGIOTIS CHATZINIKOLAOU [1] (Member, IEEE), KAROLOS-ALEXANDROS TSAKALOS [1], KONSTANTINOS RALLIS [1] (Member, IEEE), IOSIF-ANGELOS FYRIGOS [1] (Graduate Student Member, IEEE), VASILEIOS NTINAS [3] (Member, IEEE), ANTONIO RUBIO [2] (Life Senior Member, IEEE), AND GEORGIOS CH. SIRAKOULIS [1] (Member, IEEE)

[1]Department of Electrical and Computer Engineering, Democritus University of Thrace, 67100 Xanthi, Greece
[2]Department of Electronic Engineering, Universitat Politècnica de Catalunya, 08034 Barcelona, Spain
[3]Department of Electronic Systems, Aalborg University, 2450 Copenhagen, Denmark

CORRESPONDING AUTHOR: IOANNIS K. CHATZIPASCHALIS (email: ichatzip@ee.duth.gr).

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ABSTRACT Memristors and crossbar arrays are increasingly regarded as fundamental nanotechnology components for future computing and storage technologies, with promising applications in neuromorphic systems, non-volatile memories, and in-memory processing. However, their characterization and programming require precise waveform generation and reproducible signal control, which pose non-trivial challenges in experimental workflows. Developing dedicated software for waveform design in this context is particularly demanding, as it must support diverse signal types, customizable timing, and the coordination of row/column activations in crossbar architectures, while remaining intuitive for non-specialist users. This paper presents WaCPro, an open-source application that integrates waveform generation, crossbar mapping, visualization, and export functionalities into a single platform for the characterization and programming of nanoscale memristive devices and crossbar arrays. Implemented in MATLAB with a modular architecture and a graphical user interface, WaCPro enables the design and export of precisely-timed waveforms essential for the electrical stimulation of nanodevices. Export functions produce simulation- and instrumentation-ready files in widely used formats, facilitating integration into laboratory workflows, highlighting the tool's ability to bridge theory and experiment. Validation experiments demonstrate excellent waveform replication accuracy in both amplitude and timing, confirming the reliability of the proposed tool for nanoscale testing environments.

INDEX TERMS Open-source software, signal creation tool, waveform generation, memristor, crossbar arrays, neuromorphic computing.

I. INTRODUCTION

Memristive nanoelectronic devices are two-terminal electrical components first theorized by L. Chua in 1971 [1], and then experimentally materialized in 2008 at HP Labs [2]. Their main asset is that they can modify their conductance in response to voltage application, allowing switching between resistance states. They have emerged as one of the most promising candidates for next-generation nanoelectronics,

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owing to their unique resistance switching behavior and potential applications in non-volatile memory, energy-efficient neuromorphic systems, and in-memory computing [3], [4], [5], [6], [7]. Moreover, as compact, low-power, and analog-tunable devices, memristors offer opportunities to overcome the von Neumann bottleneck and realize novel computing architectures [8], [9], [10], [11], [12], [13] or even simplify the design of well-established analog circuits such as operational amplifiers, digital to analog converters, and oscillators with extra features [14]. Memristive crossbar arrays further extend the in-memory computing concept by supporting high-density integration and matrix-vector multiplication operations central to machine learning accelerators or novel quantum computing hardware [15], [16], [17], [18]. However, the experimental exploration of such nanodevices requires sophisticated instrumentation and reproducible control of electrical input stimuli.

The reproducible characterization and programming of memristors critically depend on waveform generation and signal management. Switching thresholds, endurance, and stochastic variability of these nanodevices are highly sensitive to the shape, duration, amplitude, and fidelity of applied pulses that constitute the programming scheme [19], [20], [21]. Also, it should be noted that at the nanoscale, precise control of voltage and timing parameters becomes essential as a result of ionic drift, redox processes, and localized heating effects that govern memristive switching. Even subtle differences in input waveform shape or amplitude can lead to divergent nanodevice responses and significantly influence resistance modulation and endurance, making accurate waveform synthesis a prerequisite for reliable nanodevice characterization [22]. Reliable tools for waveform design are therefore essential, not only for accurate nanodevice characterization but also for establishing reproducible experimental protocols across different laboratories.

Despite this central role, accessible tools for waveform generation in memristor and crossbar research remain scarce. Commercial arbitrary waveform generators (AWGs) provide high-precision signals but are limited by their high cost and reliance on proprietary control software. In addition, vendor interfaces often do not support unconventional pulse schemes and precise row-column coordination required for crossbar programming. On the other end of the spectrum, many researchers resort to writing ad-hoc scripts in environments such as MATLAB or Python to generate waveforms. Although flexible, these scripts typically lack graphical interfaces, require significant programming expertise, and are seldom disseminated in a standardized or reproducible form. In this way, the gap between isolated script-based solutions and commercial hardware lock-in prevents memristor research from adopting reproducible procedures more widely.

The design of dedicated software for memristor crossbar waveform generation presents additional challenges. Unlike conventional single-channel signal generation, crossbar programming involves coordinating multiple voltages across rows and columns under strict timing constraints. Such tasks require tools that can both design and visualize complex pulse trains, allow parameter tuning, and export signals in formats compatible with professional laboratory equipment and circuit simulators. Addressing these requirements while maintaining an accessible and user-friendly interface is non-trivial.

Several existing tools partially address aspects of this problem. For example, the Memristor Discovery platform by Knowm provides an open-source hardware and cross-platform software environment for basic device experiments implemented in Java, but its functionality is limited to specific hardware integration [23]. Similarly, mNET offers a visually rich Python-based simulation environment with a Graphical User Interface (GUI) for crossbar arrays, yet it does not focus on waveform generation or simulation-ready files export [24]. While these projects demonstrate the growing interest in open tools for memristor research, none combines waveform generation, crossbar mapping, visualization, and simulation files export in a single, extensible platform.

To address these gaps, we present "WaCPro" (Waveform and Crossbar Programmer), an open-source application to support waveform generation, visualization, and crossbar programming in nanotechnology research. WaCPro provides a built-in GUI based on a modular MATLAB architecture, enabling users to design step, ramp, sinusoidal, Gaussian, and custom signals, assign voltage values to the rows and columns of crossbar arrays, and export simulation-ready files in common formats such as `.csv` and `.txt`. In this way, one can use these files without any extra processing as inputs to their software (e.g., Python) or hardware (e.g., Cadence Virtuoso) designs. Last but not least, WaCPro accelerates nanoscale device testing and characterization by saving time to researchers as it produces waveform files ready to be uploaded to laboratory instruments, enhancing the reproducibility across laboratories, and thus establishing a flexible foundation for future extensions in nanotechnology research.

II. SOFTWARE ARCHITECTURE AND IMPLEMENTATION

WaCPro has been developed as a lightweight yet versatile application that unifies waveform generation, crossbar programming, visualization, and export functionalities within a single environment. The complete source code is openly distributed, ensuring transparency and full reproducibility. No proprietary or closed-source toolboxes are required, as the implementation relies solely on native MATLAB functions. The application is provided both as a standalone executable (`WaCPro.p`) and as source code, accessible via the GitHub link listed in Appendix A. MATLAB was chosen for its robust native support for instrument communication, enabling seamless integration with laboratory equipment. Nonetheless, the software's modular and platform-agnostic architecture facilitates straightforward translation to other open-source ecosystems such as Python, using PyVISA for hardware communication and PyQt for GUI development. This section presents a detailed overview of the software covering its architecture, user interface, functionalities, and design philosophy with the aim of demonstrating how its modular structure and intuitive interface support accessibility for new users while offering extensibility for advanced research applications.

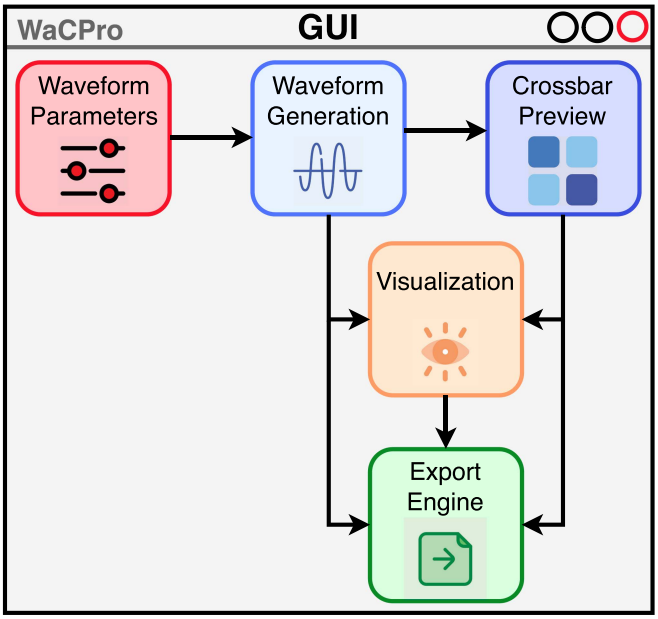


FIG. 1. High-level architecture of WaCPro. The graphical user interface (GUI) interacts with user-defined waveform parameters and coordinates the independent modules of the application.

A. SOFTWARE ARCHITECTURE

The application is implemented in MATLAB R2025a (The MathWorks, Inc., Natick, MA, USA), taking advantage of MATLAB's extensive numerical capabilities for accurate signal generation. The architecture follows a modular design that separates user interaction from core computational modules simplifying the user experience. Also, this design ensures that each functionality can be updated or extended independently, without requiring major code refactoring.

At the highest level, WaCPro consists of four major subsystems: waveform generation, crossbar preview, visualization, and the export engine. These subsystems are coordinated by the main GUI, which acts as the central control unit as seen in Fig. 1 that illustrates the overall architecture. Users interact exclusively with the GUI, which collects input parameters and dispatches them to the appropriate computational modules. These modules generate numerical signals and metadata structures that are simultaneously passed to visualization routines for inspection and to the export engine for conversion to formats compatible with third-party tools. The modular separation improves maintainability, simplifies the addition of new features, and allows reuse of core functions as standalone scripts for advanced users who prefer command-line workflows.

It is worth noting that waveforms are represented as sampled numerical arrays defined by three fundamental parameters: amplitude, duration, and time step. Additional metadata, such as waveform type and assigned crossbar row or column, are also utilized for the generation of the final waveform. This internal representation guarantees compatibility with both experimental hardware and simulation environments, while allowing signals to be scaled and adapted as needed.

B. GRAPHICAL USER INTERFACE (GUI)

The GUI is organized into three dedicated tabs, with two of them corresponding to the main function of the proposed tool and the third acting as a help tab. This tabbed structure mirrors the logical sequence followed in typical circuit simulation tasks, reducing the learning curve for new users. Initially, users can create simple waveforms to test memristor models and then proceed to the creation of 1T1R crossbar writing schemes. The tabs are the following:

- *Waveform Editor:* Provides access to built-in waveform types, including step, ramp, sinusoidal, Gaussian, and user-defined functions. Each waveform can be interactively parameterized, with real-time preview plots updating as values are adjusted. This design allows users to iteratively refine signals until they meet the desired specifications. For custom use cases, users can manually insert specific waveform parts to generate any form of arbitrary signals or load a sequence from a predefined or custom template.
- *1T1R Crossbar Writer:* Enables the assignment of specific voltage values representing weights, that correspond to specific device conductance states, onto the rows and columns of memristive crossbar arrays. Users can configure programming protocols by selecting between different writing functions and specifying pulse timings. The interface also generates timing diagrams that provide immediate feedback on signal sequencing and potential conflicts.
- *Help & Instructions Tab:* Provides embedded documentation, ensuring that users can access guidance without referring to external manuals. The tab is organized into two main parts. Section A introduces the "Waveform Editor", describing how to configure the time grid, base voltage, and segment-based waveform construction. Section B documents the "1T1R Crossbar Writer", covering the configuration of crossbar dimensions, voltage assignment, and control of timing parameters (total duration, pulse width, spacing, and delays). It explains the generation of row and column pulses, visualization of row signals and heatmaps, and export of complete datasets to timestamped `.zip` archives containing waveform files and metadata. Additional tips clarify parameter units, synchronization options, and writing scheme preview features.

A schematic overview of the GUI layout is shown in Fig. 2, where the separation of tabs is highlighted in an orange box. Moreover, note that a red "Reset App" button allows users to fully reset the application to its initial conditions, so they do not need to close it and run it again.

C. DETAILED FUNCTIONALITIES

The functionalities of WaCPro can be analyzed as follows:

1) *Waveform customization:* Using the first tab, namely "Waveform Editor", users can generate various waveforms by selecting signal type and adjusting key parameters such as amplitude, frequency, and

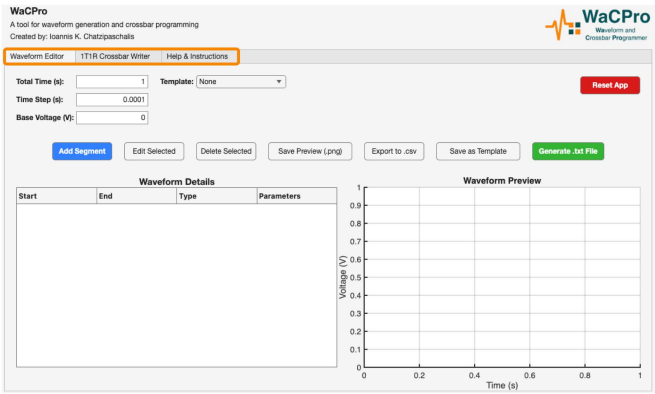


FIG. 2. Screenshot of the WaCPro graphical user interface (GUI). The tabbed layout provides dedicated interfaces for waveform editing, crossbar programming, and integrated help.

spacing. This flexibility allows the creation of both simple test pulses and complex multi-phase stimuli. There are multiple templates to choose from, namely pulse train, ramp sweep, sinusoid burst, Poisson spikes, Poisson burst train, random amplitude spikes, and a custom template in which users can manually design a waveform by adding multiple segments and then save it and load it anytime. Notably, each template opens a dedicated customization window that displays the corresponding waveform specifications.

In the waveform details window, one can see and edit the segments that comprise the produced waveform, and next to it there is a preview of it. One can save the plotting preview as an image or generate export files.

2) *Crossbar programming:* Mapping routines allow flexible assignment of signals to rows and columns of crossbar arrays. Timing synchronization is managed automatically by the proposed software tool, while advanced users can override defaults to implement custom programming protocols specifying the spacing, the delay of the pulses, or even change the type of the writing waveform among step, ramp, and half-sine. Also, there are alternative choices applied with buttons for the generation of random voltages applied to the crossbar rows as a lookup table ("Random Fill") or for the reset of the table ("Reset Table"). Column voltages are applied as pulses to ensure proper transistor switching, and their specifications can be adjusted by the user.

   One can also rename the axes legends of the crossbar plot and save it as an image. The writing lookup tables can be stored and loaded anytime, or they can be pasted from the clipboard.

   Last but not least, there is a "Preview Pulses" button that provides both a heatmap of the applied row pulses and transient plots of row and column voltages with the capability of saving them as images. These plots serve as a form of error-checking by allowing users to confirm that the signals match their intended design before exporting.

3) *File Export:* The export engine produces standardized output files that can be directly imported into laboratory equipment or simulation workflows. Exported crossbar files are annotated with metadata (e.g., sample rate, voltage scaling, activation scheme), improving traceability and reproducibility. The produced `.zip` files are named with a date and time footprint to help users distinguish different versions of their exports. Finally, the total programming energy is calculated with the export of waveform files. In this way, the tentative users can modify their writing techniques by fine-tuning the parameters of the waveform to achieve a low-power scheme, thereby decreasing the overall energy consumption of their nanocircuits.

D. DESIGN PHILOSOPHY AND USER EXPERIENCE

The development of WaCPro has been guided by three core principles that prioritize the user's needs, goals, and behaviors throughout the entire User Experience (UX) design process:

- *Usability:* An intuitive GUI reduces reliance on programming expertise, making the proposed tool easily accessible to both students and advanced researchers.
- *Reproducibility:* Standardized export formats and embedded visualization minimize inconsistencies across multiple exports and laboratories and promote reproducible workflows.
- *Extensibility:* A modular implementation enables the integration of new waveform types, visualization modes, and export formats without disrupting existing code. All core functions can also operate as standalone MATLAB scripts for advanced users who prefer command-line workflows.

The waveform precision and synchronization routines in WaCPro were designed to meet the stringent timing and amplitude requirements of nanoscale electronic devices, where nanosecond-level jitter and millivolt deviations can alter device response.

III. ILLUSTRATIVE EXAMPLES

To demonstrate the workflow and practical use of WaCPro, this section presents two proof of concept case studies: the generation of a complex waveform and the generation of write pulses for a 4×4 memristive crossbar array. These examples highlight the sequence of tasks carried out within the software tool, from waveform definition to visualization and export, without imposing any limitations to the under study crossbars.

A. WAVEFORM DESIGN

Firstly, by selecting the "Waveform Editor" tab, it is feasible to define the signal type. To demonstrate the tool's capabilities, a representative composite waveform has been generated combining a Poisson burst train, a ramp function, a sinusoidal, pulses of different amplitude, and a Gaussian distribution to

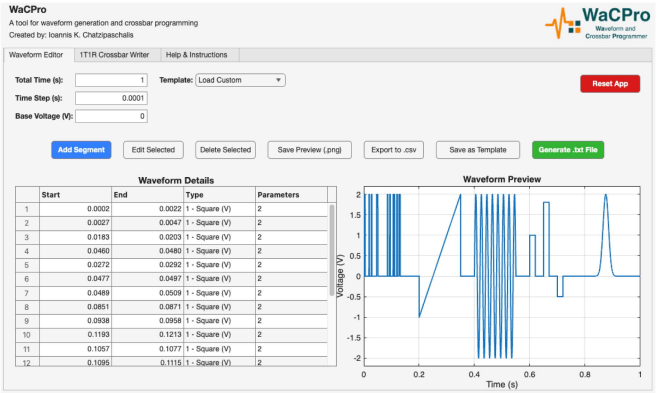


FIG. 3. Example waveform designed in the "Waveform Editor" tab. The preview plot enables verification of waveform shape and timing before export.

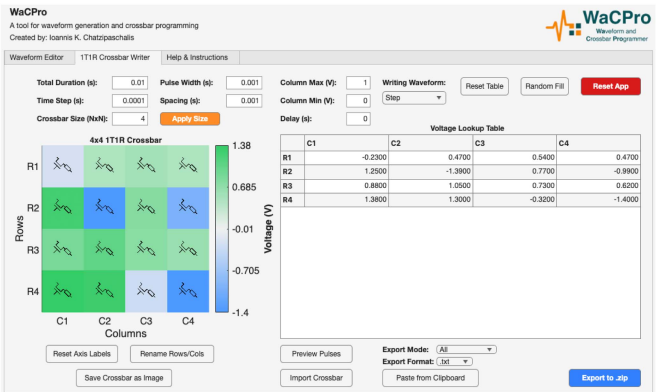


FIG. 4. Visualization of a 4 × 4 crossbar programming map showing voltage assignments across the cells.

emulate electrical dynamics encountered in nanoscale circuit applications. The preview panel (Fig. 3) provides immediate feedback, allowing verification of waveform shape and timing prior to export, while the waveform details sub-window provides information about the parameters of each waveform segment, and one can edit them by modifying the start and end time of the segment, the waveform type, and its parameters.

B. CROSSBAR PROGRAMMING

Apart from this operation, WaCPro is capable of generating crossbar writing schemes. In the proposed simple example, a voltage look-up table is mapped to the rows of a 4 × 4 crossbar using the "1T1R Crossbar Writer" tab. Users can specify the voltage levels for each row or use random generation of them, configure column control pulses, and adjust timing characteristics to achieve the desired programming sequence as seen in Fig. 4. This configuration closely resembles the biasing schemes used in memristive crossbars, where row-select and column-write voltages are applied sequentially for cell-specific programming. WaCPro provides a timing diagram and a heatmap view (Fig. 5) that illustrate how voltages are applied across the array during the write cycle.

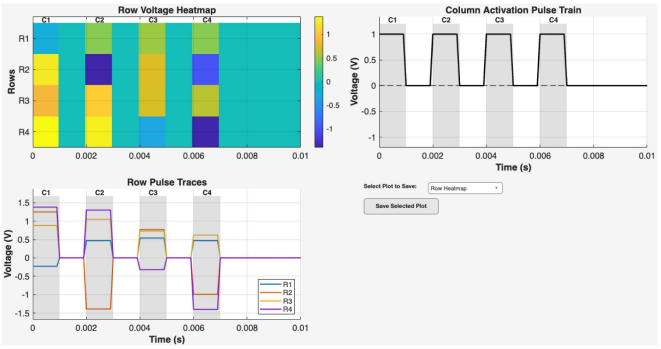


FIG. 5. Timing diagrams of row and column voltages and the corresponding heatmap view for a 4 × 4 crossbar array. These visualizations confirm correct temporal alignment between pulses.

For these signals to be exported, the user selects the desired format (`.csv` or `.txt`), and WaCPro generates a timestamped archive containing individual files for each row and column, along with metadata in a log file, where the time-voltage pairs are ready to be imported into a circuit simulator or to be fed into the provided script in Appendix A (`WaCPro_to_AWG.m`) for upload to an arbitrary waveform generator (AWG) without further processing. Also, a pop-up window appears that confirms the process, providing information about the total programming energy of the crossbar as well.

These examples demonstrate how WaCPro bridges the gap between waveform design and practical experimentation. By combining intuitive design, real-time visualization, and hardware-ready export, the software simplifies the otherwise complex task of configuring signals for memristive crossbar programming.

IV. HARDWARE VALIDATION AND NANOSCALE EXPERIMENTAL RELEVANCE

To validate the functionality of WaCPro-generated waveforms on laboratory hardware, an experimental setup comprising an Agilent 81150 A AWG, an Agilent 33220 A AWG, and an Agilent DSO90804 A digital storage oscilloscope has been implemented. The connection between MATLAB and the instruments was established via a Local Area Network (LAN), chosen over Universal Serial Bus (USB) and General Purpose Interface Bus (GPIB) for its higher data throughput since these interfaces are less suited for continuous, high-speed waveform uploads in nanoscale test loops. Waveform data were exported from WaCPro in `.csv` format and subsequently uploaded to the 81150 A AWG using a MATLAB-based Standard Commands for Programmable Instruments (SCPI) control script, available through Appendix A, transmitted over Transmission Control Protocol/Internet Protocol (TCP/IP) sockets. Although developed and tested on an Agilent 81150 A, the MATLAB control script is fully based on SCPI over TCP/IP and can be adapted to other waveform generators (e.g., Keysight, Tektronix, or Rohde & Schwarz) by adjusting command syntax for waveform definition and

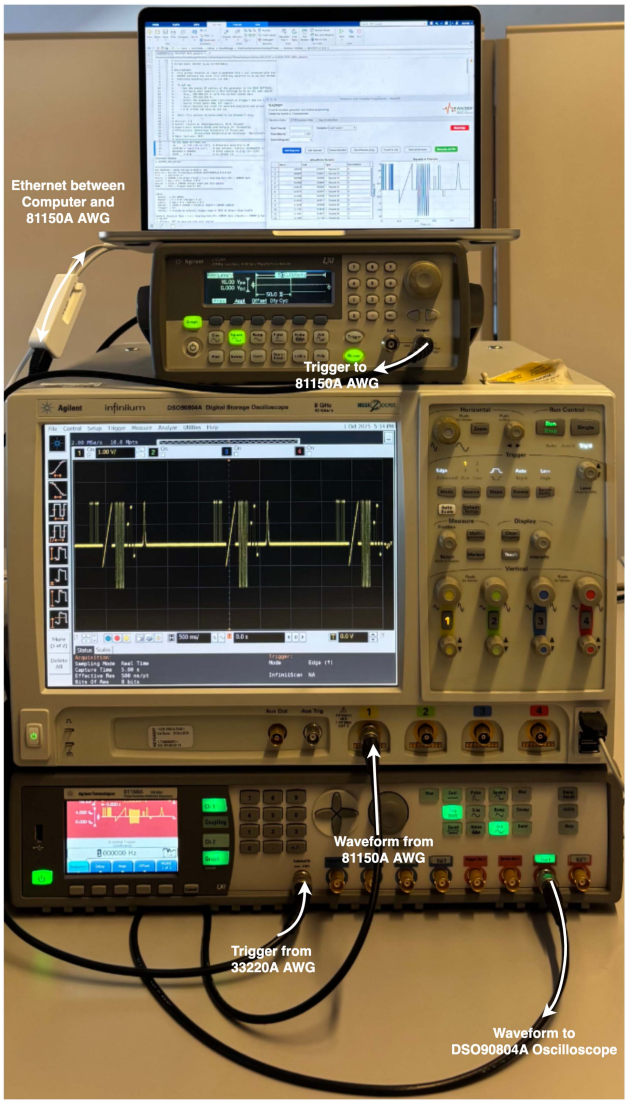


FIG. 6. Experimental setup for waveform validation. A computer running WaCPro generates the waveform and uploads it to the Agilent 81150 A AWG via LAN. The Agilent 33220 A AWG provides external trigger signals, while the Agilent DSO90804 A oscilloscope monitors the output. Connections: `Out 1` of the 81150 A to `Channel 1` of the oscilloscope, the 33220 A output to the `External IN` of the 81150 A, and computer is connected via an Ethernet cable to the 81150 A.

triggering. In addition, the procedure replicates the timing and amplitude control required for nanoscale devices such as memristor characterization, where pulse precision directly impacts switching reproducibility. The test-bench is shown in Fig. 6.

A. WAVEFORM UPLOAD AND TRIGGERING

Waveform uploads were performed using the volatile memory of the Agilent 81150 A. In this workflow, the waveform segment has been defined and prepared through the `:TRAC:DEF` and `:TRAC:SEL` commands, while the digitized waveform values have been transferred as digital-to-analog converter (DAC) integer codes using `:DATA:DAC`. Subsequently, the arbitrary waveform mode has been enabled via `:FUNC USER`, which allowed the generator to directly interpret the uploaded segment as the playback source. This method ensures rapid reconfiguration without storing waveforms permanently, making it particularly well-suited for iterative testing with different stimuli exported from WaCPro. The use of volatile memory has been deliberately chosen over non-volatile storage because it avoids overwriting previously saved waveforms in the instrument and offers faster upload performance for repeated experiments.

To guarantee accurate reproduction of the original waveform dynamics, the sample rate (SR) has not been set arbitrarily but instead derived from the temporal span of the WaCPro-exported file. More specifically, for a waveform that covers a duration of T_{csv} seconds with N resampled points, the sampling rate has been defined as $SR = N/T_{csv}$. This approach preserved the intended real-time duration of the waveform regardless of the resampling resolution.

On the amplitude side, the voltage samples have been first normalized to the unit range ($-1 \leq v \leq 1$) and then mapped into the AWG's internal DAC integer range (± 8191), which corresponds to its 14-bit resolution. The 81150A's 14-bit DAC and sub-nanosecond timing resolution make it suitable for replicating nanoscale pulse dynamics typically used in resistive switching and neuromorphic device experiments. The peak-to-peak voltage (V_{pp}) applied at the channel output has been automatically calculated from the maximum amplitude in the uploaded waveform, ensuring one-to-one scaling between the simulated signal (WaCPro) and the hardware output (AWG). In this way, both time fidelity and amplitude fidelity have been maintained during the transfer, allowing for direct correlation between simulation and experimental validation.

Apart from that, it is noted that two playback modes have been implemented in MATLAB:

- *Continuous mode:* The waveform loops indefinitely after upload, suitable for steady-state validation.
- *Triggered mode:* Playback is armed for external, or manual triggers. The first option allows discrete emission of a waveform cycle, every time an external triggering signal is present in the `External IN` bayonet Neill-Concelman (BNC) connector of the AWG, while with the latter one an output is initiated manually with the press of the `Man` key on the generator.

For our example, we have used the triggered mode with an external trigger. For the production of the triggering signal, a second AWG (Agilent 33220 A) has been used, providing a square pulse in the `External IN` BNC connector of the Agilent 81150 A. The characteristics of this triggering signal are $V_{pp-trig} = 10$ V, $f_{trig} = 0.5$ Hz, and $Duty_Cycle_{trig} = 50\%$. This signal produces multiple replicas of the WaCPro-exported waveform every 2 seconds, following the positive slope of the triggering pulse, as seen in Fig. 6.

In order to validate the measured output waveform, one can notice Fig. 7, which confirms excellent agreement between the WaCPro-designed signal (seen in Fig. 3) and the actual AWG output captured on the oscilloscope. In the oscilloscope

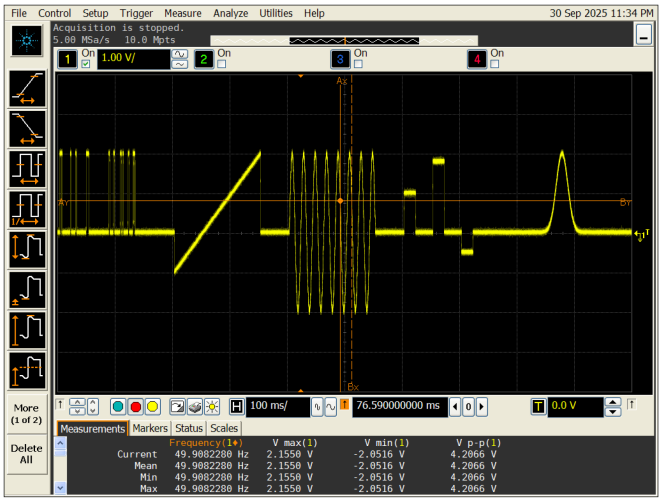


FIG. 7. Oscilloscope capture of the AWG output showing excellent agreement in waveform shape, amplitude, and timing with the WaCPro-generated signal shown in Fig. 3. Time base: 100 ms/div; amplitude: 1 V/div.

screenshot, the horizontal scale is set to 100 ms per division and the vertical scale to 1 V per division. Given that the waveform extends across ten horizontal divisions, the total duration of the signal corresponds to 1 s. Additional measurements were performed by oscilloscope software, showing the minimum and maximum voltages and the frequency of the sinusoidal part of the waveform ($V_{\min} = -2.0516$ V, $V_{\max} = 2.1550$ V, and $f_{\sin} = 49.9082$ Hz). All these values are in close agreement with the theoretical ones ($V_{\min-\text{th}} = -2$ V, $V_{\max-\text{th}} = 2$ V, and $f_{\sin-\text{th}} = 50$ Hz), thus proving identical replication.

Moreover, crossbar programming can be achieved through a straightforward approach. This would require two synchronized AWGs: one dedicated to generating row-select pulses and the other to delivering column-driving waveforms. Synchronization can be established either via a shared reference clock or by using an external trigger input, thereby ensuring precise alignment between row and column activations. In this way, the row and column waveform files exported by WaCPro could be directly mapped onto the two instruments, enabling full crossbar programming functionality.

Finally, the successful reproduction of WaCPro waveforms on laboratory equipment demonstrates the direct applicability of the software for real-world pulse generation, and the current methodology provides a solid foundation for extending WaCPro into hardware-in-the-loop testing environments.

V. DISCUSSION AND APPLICATIONS IN NANOTECHNOLOGY

WaCPro has been developed to address the need for accessible, reproducible, and flexible waveform generation in the context of nanoelectronic devices and circuits such as memristors and crossbar arrays, where precise waveform control is necessary for resistive switching, endurance cycling, and multi-level state programming. By integrating waveform design, crossbar programming schemes, real-time visualization, and export capabilities into a single platform, the software lowers technical barriers and promotes more standardized experimental workflows. Its modular MATLAB implementation ensures that the proposed tool can be extended by the community to support additional waveforms or export formats.

The impact of WaCPro can be considered along three dimensions. For nanotechnology research, it provides a reproducible and transparent workflow that reduces the likelihood of programming errors and facilitates data sharing across laboratories. For education, its intuitive GUI makes it suitable for teaching laboratories, training courses, and student projects. For dissemination, its open-source distribution and dedicated branding reinforce its identity as a software package, encouraging long-term adoption and community-driven development. In summary, it bridges the gap between waveform design and practical experimentation and contributes to advancing both research and training in emerging technologies.

Future extensions will include hardware-in-the-loop integration and coupling with nanoscale measurement environments, further enhancing reproducibility and accelerating the development of next-generation neuromorphic and in-memory nanocomputing circuits with automated nanoscale characterization loops.

APPENDIX A
SUPPLEMENTARY OPEN-SOURCE ONLINE MATERIAL

The full MATLAB source code for the WaCPro application itself (`WaCPro.m`), the executable file `WaCPro.p`, and also the script for waveform upload and playback control (`WaCPro_to_AWG.m`) are available as open-source supplementary material at: https://github.com/DUTh-FET/WaCPro. The latter script allows the definition of various AWG vendors (Agilent/Keysight, Tektronix, and Rohde & Schwarz) as there are minor differences in the SCPI commands required for the preparation and the upload procedure to AWGs of each vendor. This approach ensures reproducibility while remaining compatible with different laboratory setups and AWGs.

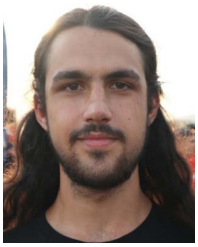
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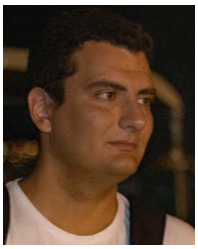
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IOANNIS K. CHATZIPASCHALIS (Student Member, IEEE) received the M.Eng. degree (Hons.) in electrical and computer engineering from the Democritus University of Thrace (DUTh), Xanthi, Greece in 2022. He is currently working toward the Ph.D. degree in ECE from DUTh, and in electronic engineering from Universitat Politècnica de Catalunya (UPC), Barcelona, Spain in the area of nanoelectronic computing circuits with learning features, under the co-supervision of Prof. Rubio and Prof. Sirakoulis. He has authored or coauthored more than 20 scientific papers in peer-reviewed journals and conferences, while has been author of one book chapter. His research interests include neuromorphic, bio-inspired, and emerging circuits and architectures, unconventional computing, and bioengineering.



PANTELIS FRAIDAKIS received the D.Eng. degree in 2025 in electrical and computer engineering from the Democritus University of Thrace (DUTh), Xanthi, Greece, where he is currently working toward the Ph.D. degree. His research interests include in-sensor computing using memristive crossbar arrays, large-scale memristive system design, and analog system design.



GEORGIOS K. KLEITSIOTIS (Graduate Student Member, IEEE) received the M.Eng. and M.Sc. degrees in engineering physics and in microsystems and nanodevices from the National Technical University of Athens (NTUA), Athens, Greece, in 2020 and 2022, respectively. He is currently working toward the Ph.D. degree in "Design and Development of Novel Neuromorphic Circuits and Systems" with the Department of Electrical and Computer Engineering, Democritus University of Thrace (DUTh), Xanthi. He has coauthored more than ten scientific works in international journals and conferences. His research interests include memristive devices and circuits, neuromorphic and unconventional computing, physics-based device modelling, and bio-hybrid electronic systems.



IOANNIS TOMPRIS (Student Member, IEEE) received the D.Eng. degree in 2023 in electrical and computer engineering from the Democritus University of Thrace (DUTh), Xanthi, Greece, where he is currently working toward the the Ph.D. degree with the Department of Electrical and Computer Engineering. His research interests include in-memory computing and modeling of memristor-based electronic devices and circuits.



ATHANASIOS PASSIAS (Graduate Student Member, IEEE) received the Diploma and M.Sc. degree in electrical and computer engineering from the Democritus University of Thrace (DUTh), Xanthi, Greece, in 2021 and 2023, respectively, where he is currently working toward the Ph.D. degree. He has coauthored more than 10 papers in international journals and conferences in the fields of neuromorphic engineering and agrifood electronics. His research interests include the design and development of neuromorphic circuits and systems, memristive devices for in-memory computing, and applications of intelligent sensing in precision agriculture.



EMMANOUIL STAVROULAKIS (Member, IEEE) received the D.Eng. degree in electrical and computer engineering and the M.Sc. degree from the Democritus University of Thrace (DUTh), Xanthi, Greece, in 2019 and 2022, respectively. He is currently working toward the Ph.D. degree from the Democritus University of Thrace (DUTh), Xanthi, Greece,. He has authored more than 10 technical papers in peer-reviewed journals and conferences. His research interests include the modeling of resistive random-access memory (RRAM) devices and neuromorphic, membrane computing using memristor crossbars, encompass unconventional electronic design, analog circuits, and heterogeneous and in-memory computing.



EVANGELOS TSIPAS (Graduate Student Member, IEEE) received the bachelor's and Integrated master's degree in electrical and computer engineering from the Democritus University of Thrace (DUTh), Xanthi, Greece, in 2021. He is currently working toward the Ph.D. degree focusing on nanoelectronics from (DUTh). He has participated in more than 15 scientific publications. His main research interests include modeling of memristive devices with variability, novel telecommunication applications harnessing memristors for 5G & mmWave applications, and memristor crossbar programming.



THEODOROS PANAGIOTIS CHATZINIKOLAOU (Member, IEEE) received the integrated D.Eng. and M.Eng. degrees in electrical and computer engineering from the Democritus University of Thrace (DUTh), Xanthi, Greece, in 2020. He is currently working toward the Ph.D. degree with the Electrical and Computer Engineering Department, DUTh. He has authored or coauthored more than 35 technical papers in peer-reviewed journals and conferences, and he has received the IEEE CASS Pre-doctoral Grant 2021 from the IEEE Circuits and Systems Society. His research interests include future and emerging computing systems and cutting-edge technologies, including memristive circuits, and systems and artificial intelligence technologies.



KAROLOS-ALEXANDROS TSAKALOS received the Dipl.-Eng. (Hons.) and Ph.D. degrees in electrical and computer engineering from the Democritus University of Thrace (DUTh), Xanthi, Greece, in 2018 and 2024, respectively. He has coauthored more than 28 scientific papers in peer-reviewed journals and international conferences. His research interests include neuromorphic computing, memristor modeling and crossbar-based in-memory systems, unconventional computing, embedded/IoT platforms, and the development of experimental tools and software frameworks for the characterization, and programming of nanoscale memristive devices.



KONSTANTINOS RALLIS (Member, IEEE) received the Dipl.-Eng. degree in electrical and computer engineering (ECE) from the Democritus University of Thrace (DUTh), Xanthi, Greece, in 2018, and the Ph.D. degree jointly from the same Department with DUTh and the Department of Electronic Engineering, Universitat Politécnica de Catalunya (UPC), Barcelona, Spain, in 2024. He is currently a Postdoctoral Researcher with the National Centre for Scientific Research "Demokritos" (NCSRD), Athens, Greece. His research interests include nanoelectronic device modeling, nanoelectronic circuit design, and lately unconventional and quantum computing.



IOSIF-ANGELOS FYRIGOS (Graduate Student Member, IEEE) received the Dipl.-Eng. degree and the Ph.D. degree in electrical and computer engineering from the Democritus University of Thrace (DUTh), Xanthi, Greece, in 2017 and 2023, respectively. He has coauthored more than 50 scientific papers in peer-reviewed journals and international conferences. His Doctoral research focused on unconventional electronic designs based on memristor technology. His research interests include high-performance computing, memristor modeling, neuromorphic and quantum-inspired computing, and unconventional hardware architectures for intelligent and energy-efficient systems.



VASILEIOS NTINAS (Member, IEEE) received the D.Eng. and M.Sc. degrees in electrical and computer engineering (ECE) from the Democritus University of Thrace (DUTh), Xanthi, Greece, in 2015 and 2017, respectively, and the dual Ph.D. degree in ECE from DUTh, and in electronic engineering from Universitat Politècnica de Catalunya (UPC), Barcelona, Spain, in 2022. From 2021 to 2025, he was a Research Associate with the Department of Electrical and Computer Engineering, TUD | Dresden University of Technology, Dresden, Germany. He is currently an Assistant Professor with the Department of Electronic Systems, Aalborg University, Copenhagen, Denmark. He has authored or coauthored more than 90 peer-reviewed journal and conference papers. His research interests include memristor-based computing architectures for edge intelligence, focusing on modeling of memristive devices, stochastic phenomena in memristive systems, and unconventional computing paradigms based on locally interconnected dynamical networks such as cellular nonlinear networks, and cellular automata. Dr. Ntinas was the recipient of the Best Paper Award on Emerging Technologies and Applications at SMACD 2023. He is currently as the Chair-elect (Secretary) with the IEEE CASS Technical Committee on CAS Education and Outreach (CASEO) and is an Active Member of the Technical Committees CNN-MAC and Nano-Giga. He was also twice as Guest Editor with IEEE TRANSACTIONS ON NANOTECHNOLOGY in 2023 and 2024, and as a Deputy General Chair with the 2023 ACM International Conference on Nanoscale Architectures (NANOARCH'23).



ANTONIO RUBIO (Life Senior Member, IEEE) received the M.Sc. and Ph.D. degrees from the Industrial Engineering Faculty, Universitat Politécnica de Catalunya (UPC), Barcelona, Spain. He was an Associate Professor with the Electronic Engineering Department, UPC and a Professor with the Physics Department, Balearic Islands University, Palma, Spain. He is currently an Emeritus Professor of electronic technology with the Telecommunication Engineering Faculty, UPC. His research interests include VLSI design and test, device and circuit modeling, high-speed circuit design, and new emerging nanodevices, and nanoarchitectures. He was as IEEE Computer Sciences Integrity Chair, from 2020 to 2021. He is an Associate Editor for the IEEE TRANSACTIONS ON COMPUTERS, a Senior Editor of the IEEE TRANSACTIONS ON NANOTECHNOLOGY, and the Chair of IEEE CASS TC Nano-Giga.



GEORGIOS CH. SIRAKOULIS (Member, IEEE) received the Dipl. Eng. and Ph.D. degrees in electrical and computer engineering from the Democritus University of Thrace (DUTh), Xanthi, Greece, in 1996 and 2001, respectively. He is a Professor in the Department of Electrical and Computer Engineering with DUTh, and he has authored or coauthored more than 330 technical papers. He is Co-editor of eight books, Co-author of 34 book chapters and Guest Editor in 18 Special Issues. His current research interests include on complex electronic systems, future and emergent electronic devices, circuits, models and architectures, unconventional computing memristors, cellular automata, quantum cellular automata, bioinspired computation/ biocomputation and bioengineering, and modeling and simulation. He is currently a Senior Editor of IEEE TRANSACTIONS ON NANOTECHNOLOGY and Associate Editor for *IEEE Nanotechnology Magazine* and IEEE TRANSACTIONS ON AGRIFOOD ELECTRONICS. He is as the VP for Pubs of the IEEE Nanotechnology Council (NTC), and he is VP-elect for Educational Activities of the IEEE NTC. He has been the General Co-chair of IEEE NMDC 2025, IEEE CAFE 2024, CNNA 2023, TPC Co-chair of IEEE NANO 2022, and of many more conferences and workshops. He is the General Co-Chair of IEEE NMDC 2025 to be held in Xanthi.